Point Estimation

Stat 266 – Unit 8 – Video Set 1 – Class Examples

Goal: To find the best (Minimum Variance Unbiased) Estimator. ⇒ MVUE

Summary:

In general: $\theta = \underbrace{\text{target parameter}}_{\text{An unknown constant.}}$, $\hat{\theta}$ or $\hat{\theta}_n$ = point estimator

↳ a r.v. prior to sampling based on a sample of n. $(Y_1, Y_2, \ldots, Y_n)$

- $\hat{\theta}$ is said to be an **unbiased estimator** for θ if $\boxed{E\left[\hat{\theta}\right] = \theta.}$
  - If $E\left[\hat{\theta}\right] \neq \theta$, then $\hat{\theta}$ is said to be **biased** for θ.
  - The **bias** of a point estimator is $B\left[\hat{\theta}\right] = E\left[\hat{\theta}\right] - \theta$.

e.g. Parameter: $\theta = \mu_Y$
Estimator: $\hat{\theta} = \bar{Y}$
$E[\bar{Y}] = E[Y] = \mu_Y$

- The **mean square for error** (MSE) of a point estimator $\hat{\theta}$ is $MSE\left[\hat{\theta}\right] = E\left[\left(\hat{\theta} - \theta\right)^2\right]$.
  - $MSE\left[\hat{\theta}\right] = V\left[\hat{\theta}\right] + \left(B\left[\hat{\theta}\right]\right)^2$.
  - If $\hat{\theta}$ is an unbiased estimator then $MSE\left[\hat{\theta}\right] = V\left[\hat{\theta}\right]$.

$E[(\hat{\theta} - E[\hat{\theta}])^2]$

- For unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the **relative efficiency** of $\hat{\theta}_1$ to $\hat{\theta}_2$ is $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) = \dfrac{V\left(\hat{\theta}_2\right)}{V\left(\hat{\theta}_1\right)}$.
  - If $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) > 1$, then $\hat{\theta}_1$ is more efficient (better). $V[\hat{\theta}_1] < V[\hat{\theta}_2]$
  - If $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) < 1$, then $\hat{\theta}_2$ is more efficient. $V[\hat{\theta}_2] < V[\hat{\theta}_1]$

Example A:

Consider the random variable, $Y \sim Binomial(n, p)$. → population proportion of successes

Note: $Y = \sum_{i=1}^{n} X_i$, where $X_i \overset{iid}{\sim} Bernoulli(p)$. → sample proportion of successes in a sample of size n.

$\left(\hat{p} = \frac{\sum X_i}{n} = \bar{X} \right)$

a) Consider the estimator, $\hat{p} = Y / n$, for p. Is $\hat{p}$ unbiased for p?

b) What is the MSE of $\hat{p}$?

c) Use the result in part (a) to suggest an estimator for σ_Y^2. Is this estimator unbiased? If not, modify it to form an unbiased estimator.

a) $E[\hat{p}] = E[Y/n] = \frac{1}{n}\mu_Y = \frac{1}{n} np = p$ ⇒ yes.

b) $MSE(\hat{p}) = V[\hat{p}] = V[Y/n] = \frac{1}{n^2}\sigma_Y^2 = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$.

c) Parameter: $\sigma_Y^2 = np(1-p) = np - np^2$.

Since $\hat{p}$ is unbiased for p, let's consider

$\hat{\sigma}_Y^2 = n\hat{p} - n\hat{p}^2$. ⇒ is this unbiased for σ_Y^2?

$\Rightarrow \hat{\sigma}_Y^2 = n\left(\frac{Y}{n}\right) - n\left(\frac{Y}{n}\right)^2 = Y - \frac{1}{n}Y^2$

$\Rightarrow E[\hat{\sigma}_Y^2] = E[Y - \frac{1}{n}Y^2] = E[Y] - \frac{1}{n}E[Y^2]$

$= np - \frac{1}{n}(\sigma_Y^2 + \mu_Y^2) = np - \frac{1}{n}\sigma_Y^2 - \frac{1}{n}n^2p^2$

$= np(1-p) - \frac{1}{n}\sigma_Y^2 = \sigma_Y^2\left(1 - \frac{1}{n}\right) = \left(\frac{n-1}{n}\right)\sigma_Y^2 \neq \sigma_Y^2$

$\Rightarrow \frac{n}{n-1}\hat{\sigma}_Y^2 = \frac{n}{n-1}Y - \frac{1}{n-1}Y^2$ is unbiased for σ_Y^2.

Example B:

Let $Y_1,\ldots,Y_n$ denote a r.s. of iid r.v.'s with density function given by

$$f_Y(y)=\begin{cases}\frac{1}{\theta}e^{-y/\theta} & , y\geq 0\\ 0 & , \text{else}\end{cases}$$

$Y \sim Exp(\mu_Y = \theta)$. $\sigma_Y^2 = \theta^2$

a) $F_Y(y) = \int_0^y \frac{1}{\theta} e^{-t/\theta}\,dt$

$= 1 - e^{-y/\theta}$, $y \geq 0$

a) Show that $\hat{\theta}_1 = nY_{(1)}$ is unbiased for θ.

b) Show that $\hat{\theta}_2 = \bar{Y}$ is unbiased for θ.

c) Find the MSE of $\hat{\theta}_1$ and $\hat{\theta}_2$. Find the relative efficiency of $\hat{\theta}_1$ vs. $\hat{\theta}_2$.

d) Find two unbiased estimators for σ_Y^2.

$f_{Y_{(1)}}(y) = n[1 - F_Y(y)]^{n-1} f_Y(y) = \frac{n}{\theta} e^{-ny/\theta}$, $y > 0$

$Y_{(1)} \sim Exp(\mu_{Y_{(1)}} = \theta/n)$.

$E[\hat{\theta}_1] = E[nY_{(1)}] = n(\theta/n) = \theta$.

$\hat{\theta}_1 = nY_{(1)}$ is unbiased for $\mu_Y = \theta$.

b) $E[\hat{\theta}_2] = E[\bar{Y}] = E[Y] = \theta$.

c) $V[\hat{\theta}_1] = V[n Y_{(1)}] = n^2 V[Y_{(1)}] = n^2 (\theta/n)^2 = \theta^2$

$V[\hat{\theta}_2] = V[\bar{Y}] = \sigma_Y^2/n = \frac{1}{n}\theta^2.$

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)} = \frac{\theta^2/n}{\theta^2} = \frac{1}{n} < 1 \quad \text{for } n \geq 2.$$

$\Rightarrow \hat{\theta}_2$ is more efficient than $\hat{\theta}_1$.

$\bar{Y}$ " " " " $nY_{(1)}$ in estimating

$\theta = \mu_Y$.

d) Parameter: $\sigma_Y^2 = \theta^2$.

1) Since $\hat{\theta}_1 = nY_{(1)}$ is unbiased for θ, consider $\hat{\theta}_1^2$.

$\hat{\theta}_1^2 = n^2 Y_{(1)}^2$.

$\cdot\ E[n^2 Y_{(1)}^2] = n^2(\sigma_{Y_{(1)}}^2 + \mu_{Y_{(1)}}^2) = n^2\left(\left(\frac{\theta}{n}\right)^2 + \left(\frac{\theta}{n}\right)^2\right) = 2\theta^2 \neq \theta^2$

So $\frac{n^2 Y_{(1)}^2}{2}$ is unbiased for θ^2.

2) Since $\bar{Y}$ is unbiased for θ, consider $\bar{Y}^2$.

$$E[\bar{Y}^2] = V[\bar{Y}] + (E[\bar{Y}])^2$$

$$= \frac{V(Y)}{n} + (E[Y])^2 = \frac{\theta^2}{n} + \theta^2$$

$$= \left(\frac{n+1}{n}\right)\theta^2 = \left(\frac{n+1}{n}\right)\sigma_Y^2 \neq \sigma_Y^2$$

So $\frac{n}{n+1}\bar{Y}^2$ is unbiased for $\sigma_Y^2 = \theta^2$.

Example C:

Suppose $X_1, \ldots, X_n$ is a random sample of independent and identically distributed $Uniform(0, \theta)$ random variables. Consider the following estimators for θ:

$\hat{\theta}_1 = cX_{(n)}$, $\hat{\theta}_2 = cX_{(1)}$, and $\hat{\theta}_3 = c\bar{X}$ for θ, where c's are constants.

a) Find c for each estimator so that the estimators are unbiased for θ.

b) Which estimator is the most efficient?

a) $\hat{\theta}_3 = c\bar{X} \Rightarrow E[\hat{\theta}_3] = cE[\bar{X}] = c\mu_X = c \cdot \frac{\theta}{2} \Rightarrow \theta \Rightarrow c = 2$

So $\hat{\theta}_3 = 2\bar{X}$ is unbiased for θ.

($\bar{X}$ is unbiased for $\mu_X = \theta/2$).

Consider $\hat{\theta} = cX_{(k)}$:

Recall: If $Y_1, \ldots, Y_n \overset{iid}{\sim} Uniform(0,1)$ then,

$Y_{(k)} \sim Beta(\alpha = k, \beta = n-k+1)$.

$E[Y_{(k)}] = \frac{k}{n+1}$, $V[Y_{(k)}] = \frac{k(n-k+1)}{(n+1)^2(n+2)}$, $k = 1, 2, \ldots, n$.

Then $X = \theta Y \sim Uniform(0, \theta)$. $X_{(k)} = \theta Y_{(k)}$.

$E[X_{(k)}] = \theta E[Y_{(k)}] = \frac{k}{n+1}\theta \neq \theta$. So $\frac{n+1}{k}X_{(k)}$ is unbiased for θ.

$\hat{\Theta}_1 = c X_{(n)} = \frac{n+1}{n} X_{(n)}$ is unbiased for Θ

$\hat{\Theta}_2 = (n+1) X_{(1)}$ is unbiased for Θ.

($\frac{n+1}{2n} X_{(n)}$ and $\frac{(n+1)}{2} X_{(1)}$ are unbiased for μ_Y)

b) $V[\hat{\Theta}_3] = V[2\bar{X}] = 4V[\bar{X}] = 4 \cdot \frac{\sigma_X^2}{n} = \frac{4\Theta^2}{12n} = \frac{\Theta^2}{3n}$.

$$V[\hat{\Theta}_{(k)}] = V\left[\frac{n+1}{k} X_{(k)}\right] = \left(\frac{n+1}{k}\right)^2 V[\Theta Y_{(k)}]$$

$$= \left(\frac{n+1}{k}\right)^2 \Theta^2 \left(\frac{k(n-k+1)}{(n+1)^2(n+2)}\right) = \frac{n-k+1}{k(n+2)} \Theta^2.$$

$$V[\hat{\Theta}_1] = V\left[\frac{n+1}{n} X_{(n)}\right] = \frac{\Theta^2}{n(n+2)}$$

$$V[\hat{\Theta}_2] = V[(n+1) X_{(1)}] = \frac{n\Theta^2}{n+2}.$$

<u>Note:</u> If $n = 1$, all variances are $\Theta^2/3$. ✓

If $n \geq 2$, $\underbrace{V[\hat{\Theta}_1]} < V[\hat{\Theta}_3] < V[\hat{\Theta}_2]$.

So $\hat{\Theta}_1 = \frac{n+1}{n} X_{(n)}$ is the most efficient for Θ.

$\Rightarrow \frac{n+1}{2n} X_{(n)}$ is most efficient for $\mu_X = \frac{\Theta}{2}$.

$\hookrightarrow$ better than $\bar{X}$.

Example D : Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed $Exponential(\beta)$ random variables. Define $C = 3Y - 2Y^2$. Find an unbiased estimator for $\mu_C = E[C]$ as a function of $\bar{Y}$.

Parameter: $\mu_C = E[3Y - 2Y^2] = 3\mu_Y - 2E[Y^2]$

$= 3\beta - 2(\sigma_Y^2 + \mu_Y^2) = 3\beta - 2(\beta^2 + \beta^2) = 3\beta - 4\beta^2.$

Since $\bar{Y}$ is unbiased for $\mu_Y = \beta$.

Consider $3\bar{Y} - 4\bar{Y}^2$. (Is this unbiased for μ_C?)

$E[3\bar{Y} - 4\bar{Y}^2] = 3E[\bar{Y}] - 4E[\bar{Y}^2]$

$= 3\mu_Y - 4(V[\bar{Y}] + (E[\bar{Y}])^2) = 3\beta - 4\left(\frac{V(Y)}{n} + \mu_Y^2\right)$

$= 3\beta - \frac{4}{n}\beta^2 - 4\beta^2 = 3\beta - 4\left(\frac{n+1}{n}\right)\beta^2 \neq 3\beta - 4\beta^2.$

Consider $a\bar{Y} + b\bar{Y}^2$.

$E[a\bar{Y} + b\bar{Y}^2] = \cdots = a\beta + b\left(\frac{n+1}{n}\right)\beta^2 \Rightarrow 3\beta - 4\beta^2$

$\Rightarrow a = 3, \quad b\left(\frac{n+1}{n}\right) \Rightarrow -4, \quad b = \frac{-4n}{n+1}.$

So $3\bar{Y} - \frac{4n}{n+1}\bar{Y}^2$ is unbiased for $\mu_C = 3\beta - 4\beta^2$.

Example E : Suppose that waiting time, Y, at a particular restaurant follows an *Exponential* distribution with a mean of $2X$, where X is a $Uniform(0,\theta)$ random variable. Find an unbiased estimator for the unconditional variance of Y as a function of $X_{(n)}$.

Parameter: σ_Y^2

- $Y|X \sim Exponential(\beta = 2X)$, $E[Y|X] = \beta = 2X$,

  $V[Y|X] = \beta^2 = 4X^2$, $E[Y^2|X] = 2\beta^2 = 8X^2$

- $X \sim Uniform(0,\theta)$, $E[X] = \frac{\theta}{2}$, $V[X] = \frac{\theta^2}{12}$, $E[X^2] = \frac{\theta^2}{3}$.

$$\Rightarrow \sigma_Y^2 = E[V[Y|X]] + V[E[Y|X]]$$

$$= E[4X^2] + V[2X] = 4\left(\frac{\theta^2}{3}\right) + 4\left(\frac{\theta^2}{12}\right) = \frac{5}{3}\theta^2.$$

Since θ is the population maximum, $\underbrace{\text{try } \frac{5}{3}X_{(n)}^2}_{\text{Is this unbiased? If not } \Rightarrow \text{ modify it}}$.

$$f_{X_{(n)}}(x) = n[F_X(x)]^{n-1} f_X(x)$$

$$= n[x/\theta]^{n-1} \cdot \frac{1}{\theta} = \frac{n x^{n-1}}{\theta^n}, \quad 0 < x < \theta$$

$$E[X_{(n)}^2] = \int_0^\theta \frac{n x^{n+1}}{\theta^n}\,dx = \frac{n}{n+2}\theta^2$$

$$E[X_{(n)}^2] = \frac{n}{n+2}\theta^2.$$

So $$E\left[\frac{5}{3}X_{(n)}^2\right] = \frac{5}{3}\left(\frac{n}{n+2}\right)\theta^2 \neq \sigma_Y^2 = \frac{5}{3}\theta^2$$

So $\frac{5}{3}\left(\frac{n+2}{n}\right)X_{(n)}^2$ is unbiased for $\sigma_Y^2 = \frac{5}{3}\theta^2$.

Example F : Suppose $Y_1,\ldots,Y_n$ is a random sample of independent and identically distributed $Poisson(\lambda)$ random variables. Find an unbiased estimator for $\theta = P(Y=0) = e^{-\lambda}$, as a function of $\bar{Y}$.

Note: $\mu_Y = \lambda$. We know $\bar{Y}$ is unbiased for λ.

Let's consider $\hat{\theta} = e^{-\bar{Y}}$. Is $\hat{\theta}$ unbiased for $\theta = e^{-\lambda}$?

$$E[\hat{\theta}] = E[e^{-\bar{Y}}] = E[e^{-\frac{1}{n}Y_1} e^{-\frac{1}{n}Y_2} \cdots e^{-\frac{1}{n}Y_n}]$$

$$= [E[e^{-\frac{1}{n}Y}]]^n = [M_Y(-1/n)]^n = [e^{\lambda(e^{-1/n}-1)}]^n$$

$$= e^{n\lambda(e^{-1/n}-1)} = e^{-\lambda(n-ne^{-1/n})} \neq e^{-\lambda}.$$

So $\hat{\theta} = e^{-\bar{Y}}$ is biased for $\theta = e^{-\lambda}$.

$\Rightarrow$ Re-visit after Rao-Blackwell Theorem.

Let $\hat{\theta}_2 = \begin{cases} 1, & \text{if } Y_1 = 0 \\ 0, & \text{otherwise} \end{cases}$. ($Y_1$ = any single observation)

$$E[\hat{\theta}_2] = 1P(Y_1=0) + 0P(Y_1 \geq 1) = P(Y_1=0) = e^{-\lambda}.$$

So $\hat{\theta}_1$ is unbiased for $e^{-\lambda}$.